

PREETINDERJEET SINGH

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SUMMARY

Data Science undergraduate experienced in statistical modeling, predictive analytics, and explainable AI — applying Python, SQL, and machine learning to real datasets across research and industry work.

EDUCATION

Guru Nanak Dev Engineering College (GNDEC), Ludhiana

Aug 2023 – Jun 2027

B.Tech in Computer Science: 8 CGPA (Till 6th Sem)

SKILLS

Languages & Query: Python, SQL

Data Analysis & ML: Pandas, NumPy, Scikit-learn, TensorFlow, Deep Learning, Statistics, Statistical Analysis, Predictive Modeling, Feature Engineering, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Model Evaluation, Data Visualization, Technical Communication

Specialized Techniques: Time-Series Forecasting (ARIMA), Explainable AI (SHAP), Panel Regression, Random Forest, Data Mining, NLP (spaCy, Sentence-Transformers, KeyBERT)

Data & Infra: Snowflake, SQLite, Databases, Neo4j, Git, Selenium, NetworkX, Data Pipelines, ETL

EXPERIENCE

Machine Learning R&D Engineer — Lupex Space

Jun 2026 – Present

- Generate, clean, and structure time-series datasets used to train missile-classification and trajectory-prediction models
- Test and validate simulation data across runs to catch inconsistencies before model training
- Lead a team of ML R&D interns, reporting directly to the co-founder, coordinating dataset and experiment work

Contractual Developer — 3S Scholars

Aug 2026 – Present

- Designed the SQL schema and query logic for a school attendance and reporting system, working directly with the CEO
- Built CSV data ingestion, daily updates, and automated backup pipelines to keep records analysis-ready
- Implementing role-based access control and CSV export for administrative reporting

Data Science Intern / Research Fellow — Punjab AI Excellence (PAI) — Remote

Jun 2025 – Jul 2025

- Curated and cleaned a 3,000+ image dataset, correcting class imbalance across five waste categories
- Evaluated model performance using accuracy, precision, and recall, reaching 87.08% test accuracy
- Fine-tuned an EfficientNet-B0 transfer-learning model under Dr. Sandeep Singh Sandha

LEADERSHIP & INVOLVEMENT

Student Coordinator, Panel Discussions — ICCT-SD 2026 (GNDEC Ludhiana × NIT Delhi)

- Coordinated panel discussion logistics and speaker/participant interactions, and co-prepared end-to-end reports and questionnaires with faculty

PROJECTS

Explainable AI-Based Spatio-Temporal Flood Risk Intelligence — Research (Preprint)

Faculty-advised research — Lead author, methodology

- Designed the study's integrated methodology, combining panel regression, Random Forest classification, SHAP explainability, ARIMA forecasting, and spatial Flood Risk Index mapping to analyze flood-risk drivers across Indian states
- Trained a Random Forest model on 640 observations, achieving 73.4% accuracy and 0.859 ROC-AUC; used SHAP analysis to identify a nonlinear deforestation threshold (~47 ha) driving flood risk
- Contributed to constructing an integrated dataset (IMD gridded rainfall, Global Forest Watch, India Flood Inventory, Indo-Floods) as part of a collective data-collection effort

Napoléon — Intent-Aware AI Web Crawler

Python · Sentence-Transformers · spaCy · KeyBERT · Selenium · NetworkX/Pyvis

- Built an intent-aware web crawler that scores crawled pages by semantic relevance using sentence-transformer embeddings, combined with spaCy-based entity extraction and KeyBERT keyword extraction
- Used NetworkX and Pyvis to generate interactive graph visualizations of crawled content and entity relationships, alongside an automated security-scanning module

PUBLICATIONS

Adaptive Multi-UAV Coordination for Efficient Disaster Response Under Uncertainty — ICAAV 2026, Paper 090

Lead-author — DRDO-organized conference, presented Aug 2026

- Co-authored and presented a paper analyzing coordination strategies for multi-UAV disaster-response systems under uncertainty
- Extended abstract accepted for the conference's Book of Abstracts; full paper under further review for Springer proceedings

CERTIFICATIONS

Time Series Analysis (HackerEarth × IIT Guwahati, 2026) · Python Programmer Certification (Infosys Springboard)